29 A sample of a radioactive isotope with nucleon number 210 decays by alpha emission into a stable isotope with nucleon number 206. The process has a half-life of 140 days.
If all the emitted alpha-particles escape from the sample but none of the daughter product escapes, what will be the percentage decrease in the mass of the source after 280 days?

A 0.48% **B** 1.43% **C** 25% **D** 75%

Ans: B

Assume the sample initially has number N unstable nuclei of nucleon number 210.

change in mass of the source after two half-lives:

$$= -\frac{3N}{4}(210) + \frac{3N}{4}(206) = -3N$$

fractional change in the mass of the source after two half-lives:

$$= \frac{-3N}{210N} = -\frac{1}{70} = -1.43\%$$

30 In medical radiotherapy, different types of radioactive emissions are chosen based on their